CARDIAC



In patients hospitalized for at least one night after non-cardiac surgery, the overall 30-day mortality rate is 1.8% with urgent/emergent surgery being associated with at least double the risk of elective surgery. [1]

Many of these deaths are linked to cardiac complications. [1] Myocardial Injury after Noncardiac Surgery (MINS), indicated by troponin levels exceeding the 99th percentile due to myocardial ischemia without ischemic features, occurs in 12%-24% of cases. [2] MINS is associated with a postoperative 30-day mortality rate of 9.8%, compared to 1% for those without it, and increases the risk of major vascular complications such as myocardial infarction and stroke. [3,4,5]

Screening Tools

The Revised Cardiac Risk Index (RCRI) includes six factors, each worth 1 point. [6] A review of 792,740 patients showed that the RCRI has moderate ability to predict major perioperative cardiac complications. [7] For patients aged 65 or older, those aged 45-64 with significant cardiovascular disease, or those with an RCRI score ≥ 1, measure NT-proBNP or BNP before noncardiac surgery to improve risk assessment. [8]

Revised Cardiac Risk Index

RCRI SCREENING	YES	NO
1. History of ischemic heart disease	☐	☐
2. History of congestive heart failure	☐	☐
3. History of cerebrovascular disease	☐	☐
4. Use of insulin therapy for diabetes	☐	☐
5. Preoperative serum creatinine > 177 µmol/L (>2.0 mg/dL)	☐	☐
6. High-risk surgery	☐	☐

SCORING Add 1 point for every **YES**

Total RCRI Points	Risk Estimate %	95% CI for the Risk Estimate
0	3.9	2.8%-5.4%
1	6.0	4.9%-7.4%
2	10.1	8.1%-12.6%
≥3	15.0	11.1%-20.0%

Total RCRI Score and corresponding risk of myocardial infarction, cardiac arrest, or death at 30 days after noncardiac surgery.

CARDIAC ALGORITHM

Patients undergoing non-cardiac surgery requiring overnight admission

Assessment of Preoperative Risk Using RCRI

If RCRI ≥ 1, patients' age is ≥ 65 yrs, or age is 45-64 yrs with significant cardiovascular disease
- ▶ Order NT - proBNP/BNP

NT-proBNP ≥ 200 ng/L or BNP ≥ 93 mg/L

Referral for Anesthesia Consult
- ▶ Preoperative ECG
- ▶ Communicate risk of major adverse cardiac event to patient

Postoperative Monitoring
- ▶ Measure Troponin daily x 48 - 72 hrs
- ▶ Obtain EKG and troponin in PACU
- ▶ Consider in-hospital shared-care management

Communicate with Primary Care Provider for Outpatient Follow-up

Prehabilitation and Optimization Recommendations

Order NT-proBNP/BNP	⚙ Criteria: overnight admission and one of the following: RCRI ≥ 1, age ≥ 65, or age 45-64 with significant cardiovascular disease ⚙ Threshold values are different for each test (BNP or NTproBNP) for triggering postoperative monitoring with troponin
Referral for Anesthesia Consult	⚙ Perioperative risk discussion and planning ⚙ If NT-proBNP is unexpectedly high without an obvious cause, consider further evaluation, including a physical exam and echocardiogram ⚙ Communicate risk estimates with patient based on RCRI score or elevated NT-proBNP values (8% for 200-1500 pmol/mL, and 16% for > 1500 pmol/mL, risk of death is 1.4% and 4.0% respectively) [8]
Postoperative Monitoring	⚙ Measure Troponin daily x 2-3 days (should not prolong hospitalization for troponin monitoring) ⚙ If Troponin > 99th percentile, MINS management required [1] ⚙ Obtain EKG and troponin level in PACU ⚙ Consider in-hospital shared-care management
Communicate with Primary Care Provider for Outpatient Follow-up	⚙ Establish follow-up plan for MINS positive patients after discharge from hospital given their increased risk of postoperative mortality